DoD Modernization Priorities

Microelectronics

Office of the Under Secretary of Defense
Research & Engineering

DoD Modernization Priorities Alignment

- Office of the Secretary of Defense (OSD) Research & Engineering (R&E) - Microelectronics
- R&E Microelectronics Roadmap
- Trusted and Assured Microelectronics (T&AM) Program
- Microelectronics Innovation for National Security and Economic Competitiveness (MINSEC) Initiative

"We cannot expect success fighting tomorrow's conflicts with yesterday's weapons or equipment." – National Defense Strategy

OSD R&E Microelectronics Roadmap: Quantifiable Assurance

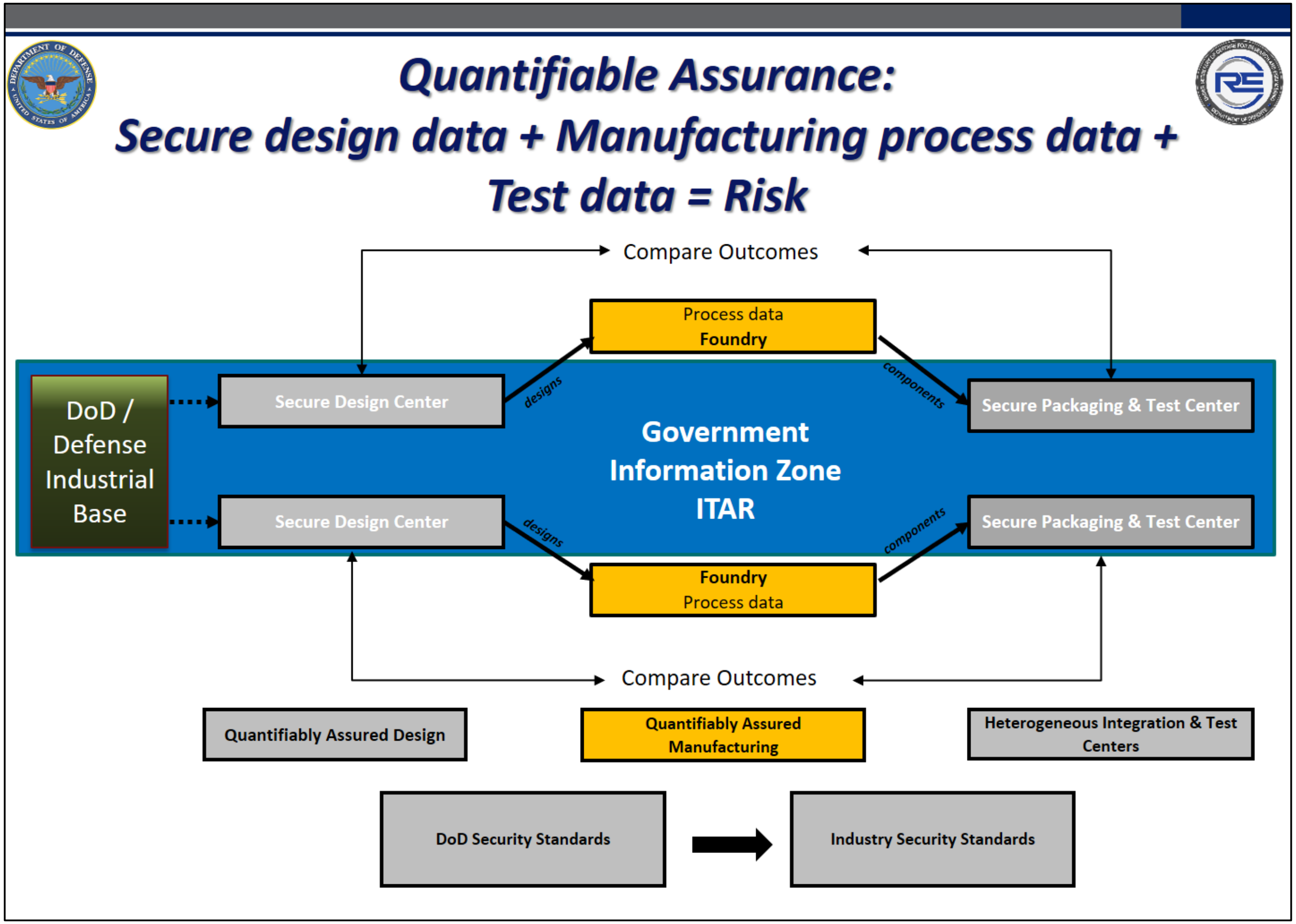
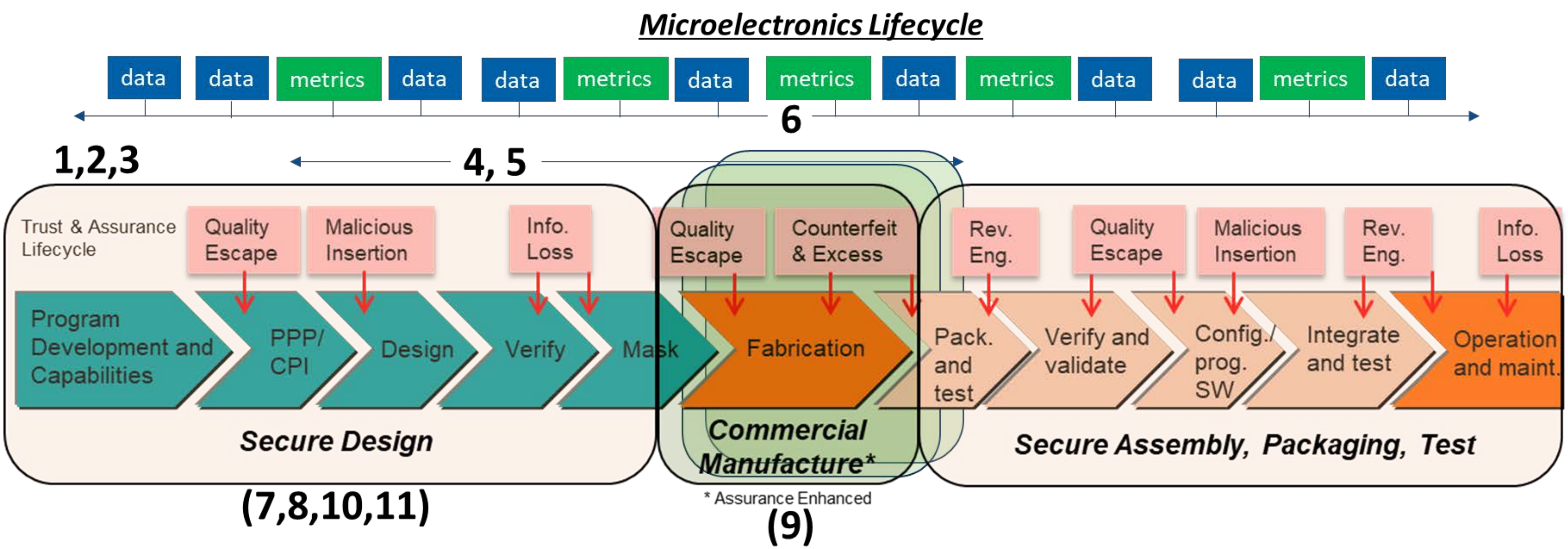


IMAGE SOURCE: Ms. Nicole Petta, "DoD Microelectronics: A Path Forward", Microelectronics Integrity Meeting, AUG 2019

DoD T&AM/MINSEC Program Project Areas

- **Access to State of the Art Commercial Technology**
  - Best practices for secure design, assembly, packaging, and test capabilities to support the defense industrial base and co-development of dual use electronics
- **Data Driven Quantifiable Assurance**
  - Full lifecycle confidentiality, integrity, verification & validation, and supply chain security for assured warfighter electronics
- **Address DoD Unique needs**
  - Develop sustainable sources of mission essential niche rad-hard electronics capabilities, and specialized Radio frequency and Eletro-Optic components
- **Maintain and Enhance U.S. Microelectronics Dominance**
  - Invigorate a secure pipeline for disruptive R&D transition, supply chain aware technology development, education and workforce

Advanced microelectronics Technology access and secure design Center - tasks



1. Develop Prototype SOTA Microelectronic Components
2. Develop Prototype SOTA Microelectronics for Obsolescence Mitigation
3. Construct System Demonstrators for Rapid Insertion
4. Develop and Evaluate Function Locking Technologies
5. Demonstrate and Evaluate Fabrication Integrity Structures
6. Construct Component Lifecycle System with Full Traceability and Provenance
7. Produce Vetted and Qualified Libraries of Design IP
8. Construct SOTA Foundry Design and Implementation Flows
9. Construct a Multi-Foundry SOTA Access Model
10. Instantiate Microelectronics Development Marketplace
11. Develop US-Based SOTA Microelectronics Persistent Expertise

DoD Research and Engineering Enterprise

Solving Problems Today – Designing Solutions for Tomorrow










DoD Research and Engineering Enterprise
https://www.CTO.mil/

Defense Innovation Marketplace
https://defenseinnovationmarketplace.dtic.mil

Twitter
@DoDCTO